

Analysis Report: Social Media Ad Campaign Performance

Submitted by: Mr. Vivek Karippara

Competition: Hackathon – Hack Studio September Edition

Track: Data Analytics – Social Media Campaign Optimization

EXECUTIVE SUMMARY

This report presents a comprehensive analysis of social media ad campaigns for TrendWave, a consumer goods company. The primary goal is to provide actionable insights to optimize ad performance, maximize ROI, and inform future budget allocations. The analysis focuses on evaluating campaign performance across different segments, identifying key trends, and providing data-driven recommendations. The findings highlight the importance of targeting specific demographics to achieve higher conversion rates and lower costs per acquisition.

INTRODUCTION

TrendWave has invested significantly in social media advertising but lacks clarity on which platforms, creatives, and audience segments are providing the best value. The key challenge is to translate raw campaign data into a clear analytical framework to guide decision-making.

Key Business Questions:

1. Which age and gender groups are responsible for the highest sales and ad spend, and how does their conversion efficiency compare?
 2. How effective is the campaign in converting impressions into clicks and clicks into sales (CTR and conversion rate)?
 3. Are there significant differences in CPA and CPM between demographic segments, and what does this imply for budget optimization?
-

DATA OVERVIEW

The dataset includes the following key variables:

- Ad ID, Campaign IDs, Age, Gender, Interest Category
- Impressions, Clicks, Spend, Total Conversion, Approved Conversion

KPIs calculated:

- **CPM** (Cost Per Mille) = (Spent / Impressions) × 1000
- **CPC** (Cost Per Click) = Spent / Clicks
- **CTR** (Click Through Rate) = (Clicks / Impressions) × 100
- **CPA** (Cost Per Acquisition) = Spent / Approved Conversion
- **CCR** (Click Conversion Rate) = (Approved Conversion/ Clicks) x 100

FINDINGS

➤ **KPI SUMMARY**

KPI	Best Segment	Worst Segment	Insight
CPM	30-34 M	45-49 F	Younger males are cost-efficient
CPC	40-44 F	35-39 M	Strong engagement in mid-age females
CTR	45-49 F	30-34 M	Creative resonance with senior females
CPA	30-34 M	45-49 F	High conversion efficiency in females
CCR	30-34 M	40-44 F	Strong conversion in younger male demographics

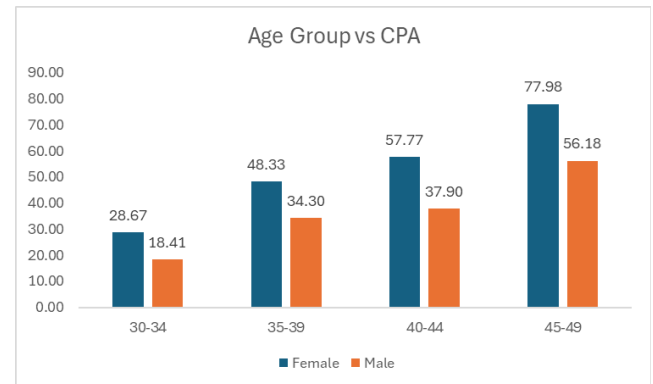
Based on the KPI summary table, the **30-34 Male** segment is the most efficient overall, delivering the lowest cost per impression (CPM) and cost per action (CPA), along with the highest conversion rate (CCR).

Conversely, the **45-49 Female** segment is the least efficient, with the highest CPM and CPA. However, this same group shows the highest click-through rate (CTR), indicating the ads are highly relevant and engaging to them, even if they are costly.

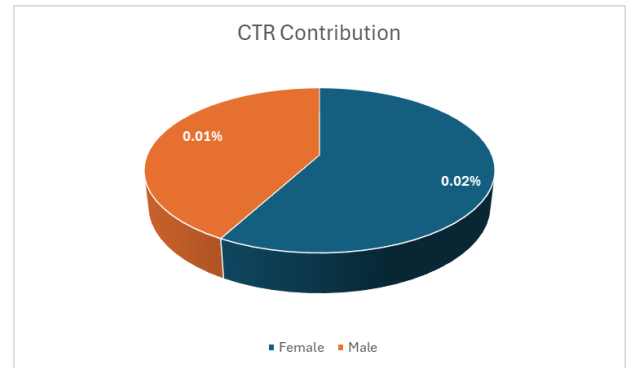
Key Insight: The best-performing audiences are younger males (30-34) for efficiency and senior females (45-49) for engagement.

➤ Audience Demographics Analysis

- **Age Group Performance:** As shown in the chart below, the **30-34 age group** had the lowest average CPA at approximately \$28.67 for females and \$18.41 for males, indicating that it is the most efficient group for generating approved conversions. In contrast, the **45-49 age group** had the highest average CPA for both genders, with the average CPA for females reaching close to \$78.



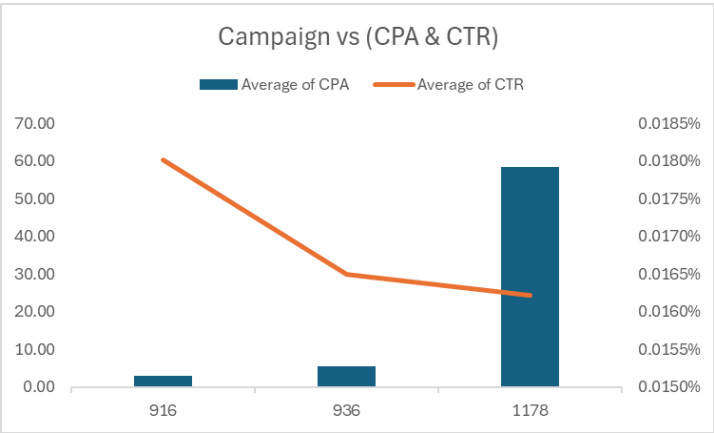
- **Gender Performance:** Male audiences had a lower average CTR (0.01%) than female audiences (0.02%). However, campaigns targeting males resulted in a significantly lower average CPA, making them a more cost-effective target. This suggests that while females are more likely to click on ads, the clicks from males lead to more affordable approved conversions.



- The 30-34 group (M and F) accounts for the most impressions and actual purchases (approved conversions).
- Females aged 45-49 have the highest spend and click volume, but conversion efficiency lags far behind the 30-34 group.
- The conversion rates and CPA rise sharply with age, indicating that older demographics are much more expensive to convert.

Age Group	Sum of Spent	Sum of Impressions	Sum of Total Conversion	Average of CPA	Average of CTR
30-34	15252.40	67993019	1431	22.95	0.012%
35-39	11112.43	42104644	626	40.61	0.016%
40-44	11589.73	39604307	523	46.82	0.020%
45-49	20750.67	63732858	684	67.62	0.022%

➤ Campaign Effectiveness



Campaign 916 has a high CTR of 0.0180% and the lowest CPA of \$3.14, making it the most cost-effective campaign. This indicates that its impressions effectively converted to clicks, and those clicks were highly efficient at generating approved conversions.

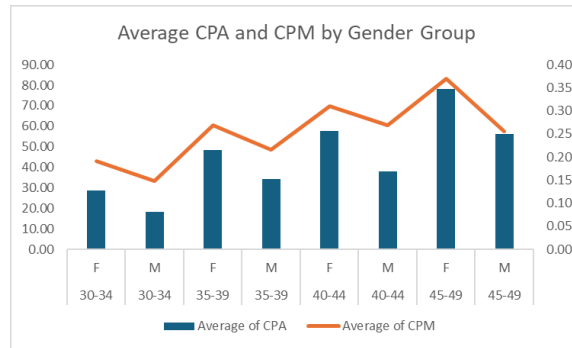
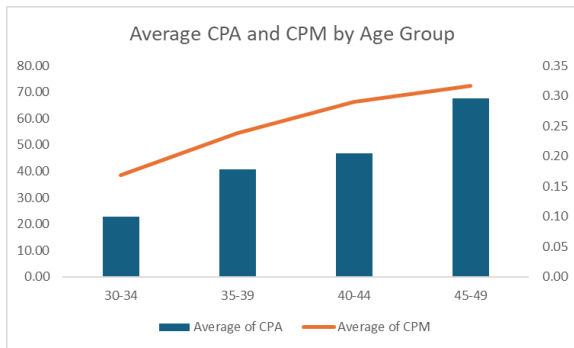
Campaign 936: This campaign shows a good balance, with a CTR of 0.0165% and a moderate CPA of \$5.50. It successfully drove a significant number of clicks and conversions at a reasonable cost.

Campaign 1178: This campaign has the lowest CTR of 0.0162% and the highest CPA of \$58.14. While it generated a large volume of impressions and clicks, the cost to achieve an approved conversion was much higher than in the other two campaigns.

Campaign	Sum of Spent	Sum of Clicks	Sum of Impressions	Average of CTR	Average of CCR
916	\$149.71	113	482925	0.0180%	16.67%
936	\$2,893.37	1984	8128187	0.0165%	13.17%
1178	\$55,662.15	36068	204823716	0.0162%	5.11%

The campaign exhibits low effectiveness in converting impressions to clicks, indicating poor initial engagement. However, the Click Conversion Rate (16.67%) is moderate, suggesting reasonable efficiency in converting clicks to sales.

➤ Cost Per Mille & CPA Analysis



- **CPM (Cost per Mille) and CPA (Cost per Action)** were the most favourable in younger to mid-aged groups (30-34, 35-39), especially for males, indicating cheaper exposure and engagement costs.
 - Segments with high CPM and CPA struggled to convert at desirable rates, suggesting a poor value proposition for ad dollars spent.
 - **CPA Variations:** CPA increases with age, ranging from \$18.41 (lowest) in the 30-34 Male segment to \$77.98 (highest) in the 45-49 Female segment—a 324% increase. Males consistently have lower CPA than females across all age groups, indicating higher conversion efficiency for male-targeted ads. Younger segments (30-34) show the best value, with CPA ~76% lower than older ones (45-49), suggesting ads perform better with younger audiences who convert at higher rates relative to spend.
 - **CPM Variations:** CPM also rises with age, from \$0.15 in 30-34 Male to \$0.37 in 45-49 Female (~147% increase). Females have higher CPM (~26% on average) than males, likely due to broader or less targeted reach in female-heavy campaigns (e.g., Campaign 936). Overall, older segments are more expensive to reach, possibly due to competitive bidding or lower engagement density.
-

RECOMMENDATIONS:

- **Increase Budgets for Top Performers:** Allocate additional spend to **30-34 Female** and **30-34 Male** segments, as they deliver robust sales at lower acquisition costs.
- **Reduce Spend on Inefficient Demographics:** Shift budgets away from **45-49 Male/Female** and **40-44 Female** categories, which combine high CPA with low sales conversion rates.
- **Refine Targeting for Costly Segments:** For groups with high spend but poor conversion, explore creative adjustments, messaging changes, or restrict exposure until KPIs improve.
- Impressions are strongly correlated with clicks ($r = 0.95$), confirming that higher visibility consistently drives engagement. However, the correlation weakens when comparing impressions to conversions ($r = 0.81$). This suggests a funnel efficiency gap — while ads attract attention, not all clicks convert into sales. Therefore, optimizing the post-click experience and refining audience targeting are key to improving ROI.
- **Increase budget allocation for Campaign ID 916.** This campaign has a proven track record of delivering the highest CTR and CCR with lowest CPA. Maintain the budget for Campaign 936, but monitor its performance closely.
- Track **CPA, CPM, CTR, and conversion rates** continuously to identify shifts in performance and adjust strategies accordingly.

CONCLUSION:

This analysis reveals clear demographic and behavioural patterns that can guide TrendWave's future ad strategy. By reallocating budget toward efficient segments and refining targeting, the company can significantly improve ROI and campaign effectiveness.